



场强 电偶极子 $\vec{E} = \frac{\vec{p}_e}{4\pi\epsilon_0 x^2}$ $p_e = ql$

直导线 ① $E_x = \frac{\lambda}{4\pi\epsilon_0 a} (\sin\theta_2 - \sin\theta_1)$ $E_y = \frac{\lambda}{4\pi\epsilon_0 a} (\cos\theta_1 - \cos\theta_2)$

② $E_x = 0$ $E_y = \frac{\lambda}{2\pi\epsilon_0 R}$ $\theta_1 = 0$ $\theta_2 = \pi$

③ $E_x = \frac{\lambda}{4\pi\epsilon_0 a}$ $E_y = \frac{\lambda}{4\pi\epsilon_0 a}$ $\theta_1 = 0$ $\theta_2 = \frac{\pi}{2}$

圆环 ① 轴线 $E = \frac{qx}{4\pi\epsilon_0 (x^2 + R^2)^{3/2}}$

② 圆心 $E = \frac{\lambda}{4\pi\epsilon_0 R} (\sin\theta_2 - \sin\theta_1)$ $E_y = \frac{\lambda}{4\pi\epsilon_0 R} (\cos\theta_1 - \cos\theta_2) \rightarrow E_x = E_y = \frac{\lambda}{4\pi\epsilon_0 R}$

圆盘 $E = \frac{\sigma}{2\epsilon_0} (1 - \frac{x}{\sqrt{x^2 + R^2}})$ $\begin{cases} x \ll R & E = \frac{\sigma}{2\epsilon_0} \\ x \gg R & \end{cases}$

球体 $\vec{E} = \frac{\rho}{3\epsilon_0} \vec{r}$

球面 $E = \begin{cases} 0 & r < R \\ \frac{Q}{4\pi\epsilon_0 r^2} & r > R \end{cases}$ 体 $E = \begin{cases} \frac{Qr}{4\pi\epsilon_0 R^3} & r < R \\ \frac{Q}{4\pi\epsilon_0 r^2} & r > R \end{cases}$

柱面 $E = \begin{cases} 0 & r < R \\ \frac{\lambda}{2\pi\epsilon_0 r} & r > R \end{cases}$ 体 $E = \begin{cases} \frac{\lambda r}{2\pi\epsilon_0 R^2} & r < R \\ \frac{\lambda}{2\pi\epsilon_0 R} & r > R \end{cases}$

电势

电偶极子 $U = \frac{\vec{p}_e \cdot \vec{r}}{4\pi\epsilon_0 r^2}$

电容 平行板 $C = \frac{\epsilon_0 S}{d}$

直导线 $U = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}$

球 $C = 4\pi\epsilon_0 R$ 半球 $C = 2\pi\epsilon_0 R$

圆筒 $U = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}$ 圆环 $U = \frac{\lambda}{4\pi\epsilon_0} \frac{2}{\sqrt{R^2 + z^2}} \Big| = \frac{\lambda\theta}{4\pi\epsilon_0 R}$

圆筒 $C = \frac{2\pi\epsilon_0 l}{\ln(R_B/R_A)}$

球面 $U = \begin{cases} \frac{Q}{4\pi\epsilon_0 r} & r < R \\ \frac{Q}{4\pi\epsilon_0 R} & r > R \end{cases}$

球 $C = 4\pi\epsilon_0 \frac{R_A R_B}{R_B - R_A}$

点 $C = \frac{Q}{U}$

电介质

电极化强度 $\vec{P} = \frac{\Sigma \vec{p}}{V} = \chi_e \epsilon_0 \vec{E}$

能量

$A_{ab} = W_a - W_b = -\Delta W$

电极化率 $\chi_e = \epsilon_r - 1$

$U_p = \int_p^0 E dl$ $\vec{E} = -\frac{dU}{dn} \vec{e}_n = -\nabla U$

束缚电荷 $b' = |\vec{P}| \cos\theta = P_n = \vec{P} \cdot \vec{e}_n$
 $= \frac{\epsilon - \epsilon_0}{\epsilon} b_0$

$W = \frac{1}{2} \Sigma q_i U_i = \frac{1}{2} CU^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} QU = \frac{1}{2} \epsilon \epsilon_0 E^2 = \frac{1}{2} \frac{D^2}{\epsilon}$

积分

$\oint_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \Sigma q_i$

自由电荷 $b = b_0 - b' = \frac{b_0}{\epsilon_r}$

$\oint_L \vec{E} \cdot d\vec{l} = 0$ $\oint_S \vec{D} \cdot d\vec{S} = \Sigma q_0$

电位移矢量 $\vec{D} = \epsilon_r \epsilon_0 \vec{E} = \epsilon \vec{E}$

$\Phi_D = \oint_S \vec{D} \cdot d\vec{S}$ $\Phi_E = \oint_S \vec{E} \cdot d\vec{S}$

$\vec{E} = \frac{E_0}{1 + \chi_e} = \frac{E_0}{\epsilon_r}$

